Analysis Notes

```r
## Make sure to run make.R first and load the necessary objects.

library(pacman)
p_load(drake, xtable, rmarkdown)
loadd(onc.dat)
loadd(cn.d.onc)
```

Is the network similarity driven by abundance of lichen?

We found support for the hypothesis that genotypically based variation in lichen network structure is potentially driven by variation in bark roughness. In an initital PerMANOVA with sequential partitioning of variance explained, genotype has a strong, statistically significant effect (R2 = 0.37, p-value = 0.040) when including tree traits and lichen community indices in the model. In this ordering, condensed tannins (R2 = 0.09, p-value = 0.486) and species richness (R2 = 0.11, p-value = 0.017) are both significant, although with low explanatory power. After moving genotype to the final predictor in the model, genotype is no longer significant and has low explanatory power, while both bark roughness (R2 = 0.14, p-value = 0.006) and species richness (R2 = 0.20, 0.002) are now significant and have increased in the variance in network similarity explained. Taken together, these results suggest that there is potentially a causal relationship between genotypically explained variance in bark roughness and lichen community species richness that affects the similarity of lichen network structure.

```r
cn.geno.trait.pc.sr <- vegan::adonis2(
                              cn.d.onc~ geno + CT + pH + CN + BR + PC + SR,
                              by = "term",
                              data = onc.dat,
                              mrank = TRUE,
                              permutations = 100000)

cn.trait.pc.sr.geno  <- vegan::adonis2(
                                cn.d.onc~ CT + pH + CN + BR + PC + SR + geno,
                                by = "term",
                                data = onc.dat,
                                mrank = TRUE,
                                permutations = 100000)
```

```r
as.data.frame(cn.geno.trait.pc.sr)
```

```r
as.data.frame(cn.trait.pc.sr.geno)
```

```r
xtable(as.data.frame(cn.geno.trait.pc.sr))
```

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	Df	SumOfSqs	R2	F	Pr(>F)
geno	9.00	257.29	0.37	2.35	0.04
CT	1.00	60.20	0.09	4.95	0.05
pH	1.00	7.82	0.01	0.64	0.45
CN	1.00	13.75	0.02	1.13	0.30
BR	1.00	14.72	0.02	1.21	0.28
PC	1.00	11.13	0.02	0.92	0.35
SR	1.00	73.32	0.11	6.03	0.02
Residual	21.00	255.22	0.37		
Total	36.00	693.44	1.00		

```r
xtable(as.data.frame(cn.trait.pc.sr.geno))
```

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	Df	SumOfSqs	R2	F	Pr(>F)
CT	1.00	42.57	0.06	3.50	0.07
pH	1.00	10.00	0.01	0.82	0.37
CN	1.00	19.52	0.03	1.61	0.21
BR	1.00	99.98	0.14	8.23	0.01
PC	1.00	29.88	0.04	2.46	0.12
SR	1.00	140.15	0.20	11.53	0.00
geno	9.00	96.14	0.14	0.88	0.56
Residual	21.00	255.22	0.37		
Total	36.00	693.44	1.00		

How does bark roughness affect the structure of the lichen networks?

Is it driven by lichen abudance (Percent Cover)?

Examination of the variaince in lichen network similarity explained by species richness and the network metrics of size and centrality, supported the hypothesis that network structure is both a function of increasing numbers of species (i.e., nodes in the network) and other structural differences. In analyzing two PerMANOVAs with these factors, we found that network size ($R2 = 0.77$, p-value < 0.001) and centrality ($R2 = 0.04$, p-value < 0.001) both explained vaiation in network similarity regardless of whether species richness was the first or last factor entered into the model. Additionally, although variation in network similarity explained by centrality was low, it was significant even after partitioning the variance explained by network size. Based on this, we can conclude that network similarity is both of function of the number of lichen species in the community determining network size and other aspects of network strcutre, such as centralization.

```r
cn.l.cen.sr  <- vegan::adonis2(
                           cn.d.onc~ L + Cen + SR,
                           by = "term",
                           data = onc.dat,
                           mrank = TRUE,
                           permutations = 100000)

cn.sr.l.cen  <- vegan::adonis2(
                           cn.d.onc~ SR + L + Cen,
                           by = "term",
                           data = onc.dat,
                           mrank = TRUE,
                           permutations = 100000)
```

```r
as.data.frame(cn.l.cen.sr)
```

```r
as.data.frame(cn.sr.l.cen)
```

```r
xtable(as.data.frame(cn.l.cen.sr))
```

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```r
xtable(as.data.frame(cn.sr.l.cen))
```

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	Df	SumOfSqs	R2	F	Pr(>F)
L	1.00	601.44	0.87	376.37	0.00
Cen	1.00	36.58	0.05	22.89	0.00
SR	1.00	2.69	0.00	1.68	0.19
Residual	33.00	52.73	0.08		
Total	36.00	693.44	1.00		

	Df	SumOfSqs	R2	F	Pr(>F)
SR	1.00	77.83	0.11	48.71	0.00
L	1.00	536.36	0.77	335.64	0.00
Cen	1.00	26.52	0.04	16.60	0.00
Residual	33.00	52.73	0.08		
Total	36.00	693.44	1.00		